

LAB4

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1.总体框架

该代码实现主要由三个部分组成;

```
patrol_points_publisher.py
car_control_node.py
car_motion_node.py
```

patrol_points_publisher实现巡逻点信息发布, car_motion_node实现

2.具体代码实现

2.1 patrol_points_publisher

- 1.节点初始化 启动一个 ROS2 节点, 并创建 StaticTransformBroadcaster 用于发送静态 TF。
- 2.巡逻点定义 每个巡逻点由三个变量组成: 名称, x 坐标, y 坐标 小车将根据这些巡逻点顺序巡逻。
- 3.发布巡逻点名称列表, 发布静态 TF 坐标通过 Static TF 发布, 仅发布一次即可, 每个巡逻点在 RViz2 中都是一个独立坐标系

2.2 car_motion_node

该节点负责模拟小车在二维平面中的运动, 并实时发布小车相对于 world 坐标系的动态变换, 该节点订阅控制节点发布的速度信息 /cmd_vel, 根据速度做积分计算位置

- 1.节点初始化 订阅话题 /cmd_vel, 获取小车的线速度
- 2.小车状态初始化

```
self.x = 0.0
self.y = 0.0
self.vx = 0.0
self.vy = 0.0
self.last_time = self.get_clock().now()
```

last_time 用于记录小车上一次的位置

- 3.订阅速度命令回调, 更新小车位置, 发布 TF (world → car) 每一次更新, 小车都会发布自己的位置信息

2.3 car_control_node

负责根据巡逻点的位置计算小车需要的速度指令, 并将速度发布给 car_motion_node。

- 1.初始化

```
self.tf_buffer = Buffer()
self.tf_listener = TransformListener(self.tf_buffer, self)
self.pub = self.create_publisher(Twist, 'cmd_vel', 10)
```

发布者 /cmd_vel: 发布 Twist 消息作为小车的控制输入 (vx, vy)

- 2.接受巡逻点列表

```
self.create_subscription(String, 'patrol_points', self.patrol_points_cb, 10)
```

将 "point_1,point_2,point_3..." 转换为列表,第一次收到巡逻点时初始化 current_idx 为 0,之后更新列表但不重置巡逻顺序

- 3.控制循环 control_loop a.获取当前目标点,从 TF 查询位置,计算与目标点的误差 b.判断是否到达目标点: 到达目标点 → 切换下一个目标

```
if dist < self.arrival_tol:
    self.current_idx = (self.current_idx + 1) % len(self.patrol_points)
    self.pub.publish(Twist()) # 停止
    return
```

c. 速度控制 小车越远速度越大, 越近速度越小

```
vx = self.kp * dx
vy = self.kp * dy
```

d.发布速度指令

```
twist = Twist()
twist.linear.x = float(vx)
twist.linear.y = float(vy)
self.pub.publish(twist)
```

3.实现结果

小车在预设定的point 1-4 之间巡逻，发布小车的位置和速度信息

```
问题 输出 调试控制台 终端 窗口
zzx@zzx-VirtualBox:~/myws$ ros2 run car patrol_p
oints_publisher
int_2,point_3,point_4
[INFO] [1763105983.830804013] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763105988.827899869] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763105993.835089279] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763105998.838334415] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106003.832304603] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106008.828370212] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106013.829641285] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106018.828000713] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106023.827659078] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106028.828020211] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[INFO] [1763106033.827439643] [patrol_points_pub
lisher]: Published patrol point list: point_1,po
int_2,point_3,point_4
[]

[INFO] [1763106036.118268879] [car_motion_node]:
car pose (2.30, 3.32)
[INFO] [1763106036.169148698] [car_motion_node]:
car pose (2.25, 3.33)
[INFO] [1763106036.218692920] [car_motion_node]:
car pose (2.20, 3.34)
[INFO] [1763106036.270071046] [car_motion_node]:
car pose (2.15, 3.34)
[INFO] [1763106036.320061206] [car_motion_node]:
car pose (2.10, 3.35)
[INFO] [1763106036.368655045] [car_motion_node]:
car pose (2.05, 3.36)
[INFO] [1763106036.419670115] [car_motion_node]:
car pose (2.00, 3.36)
[INFO] [1763106036.469575704] [car_motion_node]:
car pose (1.95, 3.37)
[INFO] [1763106036.518069346] [car_motion_node]:
car pose (1.90, 3.37)
[INFO] [1763106036.570433135] [car_motion_node]:
car pose (1.85, 3.38)
[INFO] [1763106036.618038843] [car_motion_node]:
car pose (1.80, 3.38)
[INFO] [1763106036.669126215] [car_motion_node]:
car pose (1.75, 3.39)
[INFO] [1763106036.719520062] [car_motion_node]:
car pose (1.70, 3.39)
[INFO] [1763106036.769696560] [car_motion_node]:
car pose (1.65, 3.40)
[INFO] [1763106036.818408913] [car_motion_node]:
car pose (1.60, 3.40)
[INFO] [1763106036.868212686] [car_motion_node]:
car pose (1.55, 3.41)
[INFO] [1763106036.918186799] [car_motion_node]:
car pose (1.50, 3.41)
[INFO] [1763106036.971062656] [car_motion_node]:
car pose (1.45, 3.41)
[]

[INFO] [1763106035.856611605] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.18, dis
t=2.61
[INFO] [1763106035.956428423] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.17, dis
t=2.51
[INFO] [1763106036.055512155] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.15, dis
t=2.41
[INFO] [1763106036.155742185] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.14, dis
t=2.31
[INFO] [1763106036.258014589] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.13, dis
t=2.21
[INFO] [1763106036.356478707] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.12, dis
t=2.11
[INFO] [1763106036.455866690] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.11, dis
t=2.01
[INFO] [1763106036.555458051] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.10, dis
t=1.91
[INFO] [1763106036.656504199] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.09, dis
t=1.81
[INFO] [1763106036.757524674] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.09, dis
t=1.71
[INFO] [1763106036.856506339] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.08, dis
t=1.61
[INFO] [1763106036.956459793] [car_control_node]
: Moving towards point_4: vx=-1.00, vy=0.07, dis
t=1.51
[]
```